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Extending the 3D scanning range of reflective dynamic focusing device-based laser scanners

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Reflective dynamic focusing devices (RDFDs) have shown their potential in laser scanning as high-performance laser Z-direction focusing devices. However, the scanning range of RDFD-based scanners is limited by aberrations during dynamic focusing. An aspheric symmetry correction (ASC) method was proposed to extend the effective scanning range. An aspheric lens was introduced to correct the optical path difference (OPD) and optimize aberrations. As a result, the scanning range in the three-dimensional (3D) space increased by 15.2%. The ASC method has been proven to extend the 3D scanning range of RDFD-based scanners and may have broad application prospects. © 2023 Optica Publishing Group

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Laser three-dimensional (3D) scanning technology, which relies on the rapid movement of the laser focus in the 3D space, has found widespread applications in microstructure manufacturing [1–4], laser cleaning [5–7], and large format texturing [8–11]. The X and Y positions of the laser focus are typically controlled by two-axis galvanometers [12–16], with scanning speeds of up to 10 m/s achievable at a working distance of 160 mm [17]. However, achieving comparable scanning speed for the Z position of the laser focus remains a challenge. It is a crucial factor limiting the maximum scanning speed of the laser 3D scanning system [18]. The scanning speed of the laser focus along the Z direction, when using a moving concave lens, is three orders of magnitude slower than that of the X and Y directions [18]. This is because the parallel movement speed of the concave lens lags far behind the deflection speed of two-axis galvanometers [19,20]. Methods that manipulate the shape of the lens based on the principle of electro-wetting [21,22] or shape-variable lenses [23–26] can rapidly shift the Z direction position of the laser focus. However, the surface shape of variable lenses is difficult to control precisely [27]. Reflective dynamic focusing devices (RDFDs) use two highly dynamic characteristic galvanometers to convert angular motion into axial motion, increasing the laser scanning speed in the Z direction [28,29]. However, when shifting the focus Z position, RDFDs need to break the symmetry of the optical path, which introduces additional optical path difference (OPD), affecting the focusing quality. Especially when RDFDs are used in conjunction with an XY scanner for 3D scanning, the XY scanning further amplifies the OPD. This leads to a decrease in the focusing quality and uniformity in the central area of the scanning field, and even the edge area cannot be used for processing. Consequently, the development and application of RDFDs was limited.

In this Letter, the reasons for the limited scanning range of RDFD-based 3D laser scanners were discussed. The aberrations caused by the broken symmetry of the optical path were analyzed. An aspheric symmetry correction (ASC) method was proposed to optimize OPD and extend the scanning range. An aspheric correcting lens (CL) was placed in front of the focal point of the second off-axis parabolic mirror to compensate for the OPD between the edge rays and the chief ray, and the symmetry of the optical path was improved. Etching experiments were performed to verify the effectiveness of the ASC method.

The optical path of the RDFD-based laser scanner is shown in Fig. 1(a). The RDFDs consisted of two off-axis parabolic mirrors (PM_1 and PM_2) and galvanometers (G_1 and G_2) placed at the focal points of PM_1 and PM_2, respectively. The G_1 and G_2 were rotatable. When the focus Z position was not shifted, the angle between the outgoing and incident beam of G_1 and the angle between the outgoing and incident beam of G_2 were equal to the off-axis angle θ of the parabolic mirrors. The laser-focusing Z (the direction along the optical axis) position was controlled by rotating G_1 and G_2. The rotation angles α of G_1 and G_2 were equivalent, the rotation directions were opposite, and the rotation angle of the outgoing beam was $\Delta\theta = 2\alpha$. When G_1 and G_2 rotated, the primary focus positions between PM_1 and PM_2 were shifted forward or backward, as shown by points A_1, B_1, and C_1 in Fig. 1(a). After reflection by G_2, it was scanned in two dimensions (2D) in the plane perpendicular to the Z direction by the XY scanner and focused by the F-theta lens (FL), so secondary focus positions (dynamic focusing positions) shifted forward or backward, along the Z direction after the FL, as shown by points A_2, B_2, and C_2 in Fig. 1(a), and 2D scanning can be performed by XY scanner in the Z plane corresponding to points A_2, B_2, and C_2. The focus-shift distance ΔZ can be derived as

$$\Delta Z = K(R, D) f \alpha, \tag{1}$$

where K is a factor related to RDFDs and depends on the curvature R of PM_1 and PM_2, the beam diameter D, and the

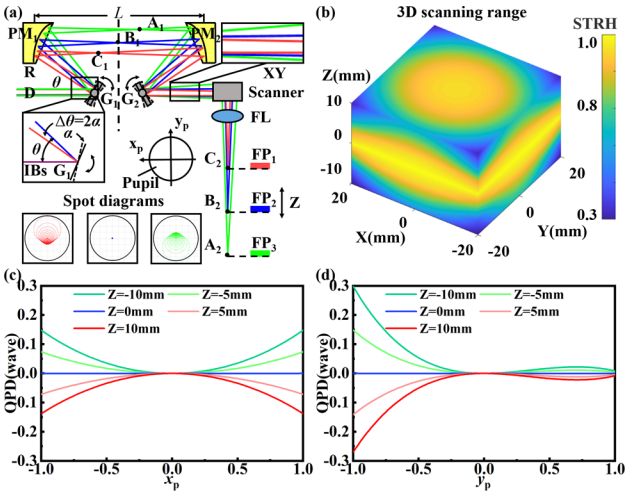


Fig. 1. Schematic diagram of the aberration generated by the RDFD-based laser scanner during the focusing process. G_1, G_2, galvanometers; PM_1, PM_2, off-axis parabolic mirrors; D, beam diameter; R, curvature of PM_1 and PM_2; L, distance of PM_1 and PM_2; XY scanner, two-axis galvanometers; FL, F-theta lens; FP_1, FP_2, FP_3, focal planes; A_1, B_1, C_1, primary focus positions; A_2, B_2, C_2, secondary focus positions; IBs, incident beams; θ, off-axis angle of PM_1 and PM_2; α, rotation angle of G_1; $\Delta\theta$, rotation angle of outgoing beam; STRH, Strehl ratio; x_p/y_p, components of the normalized entrance pupil coordinates in the sagittal/tangential directions; Z, Z position of the focus. (a) Optical path of a conventional RDFD-based laser scanner; (b) 3D scanning range of a conventional RDFD-based laser scanner; (c) and (d) Variations of OPD with x_p/y_p and Z (D = 8 mm, R = 304.8 mm, $\theta = 45°$, the focal length f of FL = 100 mm, $L = 410.729$ mm).

focal length f of the FL. In our simulations and experiments, $K = 0.033/°$.

According to the aberration theory [30] and the Abbe sine condition [31], there was no coma in the system when the distance between PM_1 and PM_2 satisfied that the primary focus position B_1 was located on the symmetry axis of PM_1 and PM_2. In this case, the optical path between G_1 and G_2 was perfectly symmetric, as shown by the solid blue line, with only a small amount (−0.017 waves, in our system) of spherical aberration. During the dynamic focusing process, G_1 and G_2 deflected (−Z, green; +Z, red) from the initial angle (Z = 0 mm, blue), and the symmetrical structure of the optical path was broken by the rotation of G_1 and G_2. As depicted by the red and green solid lines shown in the rectangular zoom in Fig. 1(a), the laser entering the XY scanner was not symmetrical. Consequently, the broken symmetry resulted in the formation of mainly coma aberrations at the focal planes FP_1 and FP_3, affecting the focus quality in spot diagrams.

Figures 1(c)–1(d) show the variations of OPD with x_p/y_p and Z, where x_p/y_p were the components of the normalized entrance pupil coordinates in the sagittal/tangential directions with larger absolute values closer to the edge of rays [30], and Z was the Z direction dynamic focusing position. As shown in Fig. 1(c), the absolute values of OPD of the ray in the sagittal direction increased from the value 0 as the Z position of the focus moved away from Z = 0 mm to Z = −10 mm/Z = 10 mm. The OPD was also increased with x_p, which indicated that the closer the ray in the sagittal plane was to the edge, the larger the OPD between it and the chief ray was. When Z = −10 mm and $x_p = \pm 1.0$, the OPD was 0.148 waves, and when Z = 10 mm and $x_p = \pm 1$, the OPD was −0.138 waves. The agreement of the values at different Z planes, for $x_p = \pm 1.0$, $x_p = \pm 0.9$, $x_p = \pm 0.8$, and $x_p = \pm 0.7$, indicated that this edge effect of rays was symmetric in the sagittal plane of the beam.

As shown in Fig. 1(d), in the tangential plane, the OPD of the ray increased to 0.298 waves when Z = −10 mm and $y_p = -1.0$, and the OPD of the ray was 0.009 waves when Z = −10 mm and $y_p = 1.0$. The OPD of the ray decreased to −0.269 waves when Z = 10 mm and $y_p = -1.0$, and the OPD of the ray was -0.008 waves when Z = 10 mm and $y_p = 1.0$. The disagreements of the values at different Z planes, for $y_p = \pm 1.0$, $y_p = \pm$ 0.9, $y_p = \pm 0.8$, and $y_p = \pm 0.7$ indicated that this edge effect of rays was asymmetric in the tangential plane of the beam. However, this asymmetry introduced additional coma aberrations, which was consistent with the spot diagrams in Fig. 1(a). The asymmetry of the beam and increased OPD was combined to introduce aberrations.

The XY scanner scanned the laser-focusing spot in X and Y directions at each Z plane. However, it acted as an aberration amplifier, further amplifying the aberration at the center of each focusing plane [8], so the axial aberration affects the 3D scanning range of the RDFD-based laser scanner. When the diffraction of the ray was considered, the Strehl ratio (STRH) reflected the effect of the OPD on the focusing effect in terms of energy [32]. Therefore, the STRH was used to evaluate the focusing effect within the scanning range, taking the value of 0 to 1, with closer to 1 indicating better focusing. A STRH of ≥0.8 is generally considered to correspond to diffraction-limited performance and effective processing. For STRH of ≤0.3, the behavior of the STRH becomes uncorrelated to the processing quality due to large aberrations [32]. The STRH, which was simulated by ZEMAX, within the 3D scanning range is shown in Fig. 1(b). The optical aperture D was 8 mm, the laser wavelength λ was 1064 nm, the curvature R of two off-axis parabolic mirrors was 304.8 mm, the focal length f of FL was 100 mm, and the distance L of PM_1 and PM_2 was 410.729 mm. When Z = 0 mm, the STRH was close to 1 within the scanning range of 40×40 mm^2. However, when the laser focus moved backward or forward, the STRH decreased rapidly. When Z = ±10 mm, the STRH at the corners of the scanning field decreased to only about 0.3, affecting the effective scanning range.

An aspheric symmetry correction (ASC) method was proposed to extend the 3D scanning range of RDFD-based laser scanners. The schematic diagram of the ASC method is shown in Fig. 2(a). An aspheric correcting lens with a thicker center and thinner sides was inserted between PM_2 and G_2 to improve the symmetry of the optical path to reduce aberrations, and the initial focused beam (Z = 0 mm, blue) passed through the center of the lens [31]. The aperture of CL and the distance from CL to G_2 were adjusted so that light within the field of view of G_2 could pass through. By adjusting the OPDC (the optical path difference referenced to the chief ray, expressed in wave number) and RAED (the angle between the interface normal and the ray of light after refraction or reflection on the surface, expressed in degrees) operands in ZEMAX so that the edge rays were optimized towards a symmetric trend when the focusing light path with asymmetry passes through the lens, the CL can optimize different focusing light paths in different face shape regions due to the design of placing the CL between PM_2 and G_2. As depicted by red and green solid lines shown in the rectangular zoom in

Table 1. Variations of OPD (Wave) with Z and x_p/y_p

CL	Z (mm)	$x_p = -1.0$	$x_p = +1.0$	$y_p = -1.0$	$y_p = +1.0$
Without	Z = −10	0.148	0.148	0.298	0.009
	Z = 10	−0.138	−0.138	−0.269	−0.008
With	Z = −10	0.004	0.004	0.088	−0.139
	Z = 10	0.033	0.033	−0.110	0.089

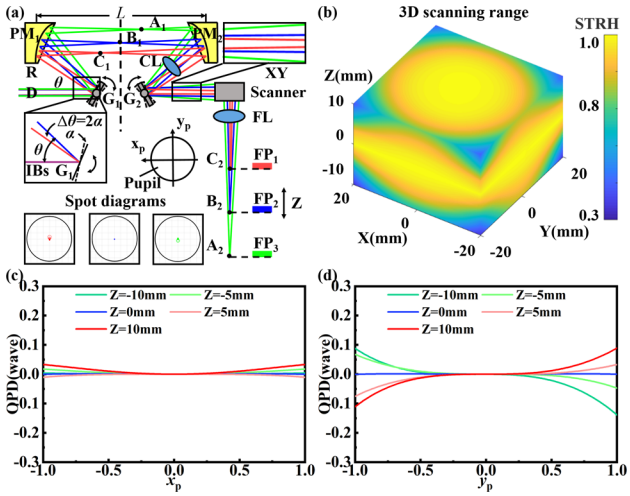


Fig. 2. Schematic diagram of the correcting lens to optimize the aberration generated by the RDFD-based laser scanner during the focusing process. G_1, G_2, galvanometers; PM_1, PM_2, off-axis parabolic mirrors; D, beam diameter; R, curvature of PM_1 and PM_2; L, distance of PM_1 and PM_2; CL, aspheric correcting lens; XY scanner, two-axis galvanometers; FL, F-theta lens; FP1, FP2, FP3, focal planes; A_1, B_1, C_1, primary focus positions; A_2, B_2, C_2, secondary focus positions; IBs, incident beams; θ, off-axis angle of PM_1 and PM_2; α, rotation angle of G_1; $\Delta\theta$, rotation angle of outgoing beam; STRH, Strehl ratio; x_p, y_p, components of the normalized entrance pupil coordinates in the sagittal/tangential directions. Z, Z position of the focus. (a) Optical path of an RDFD-based laser scanner with CL. (b) 3D scanning range of an RDFD-based laser scanner with CL. (c) and (d) Variations of OPD with x_p/y_p and Z (D = 8 mm, R = 304.8 mm, $\theta = 45°$, the focal length f of FL = 100 mm, L = 410.729 mm).

Fig. 2(a), the laser entering the XY scanner was symmetrical in the XY plane.

The optimized OPD in the sagittal plane is shown in Fig. 2(c), and the maximum OPD value was only 0.033 waves. As in the tangential plane, the OPD of the ray was 0.088 waves when Z = −10 mm and $y_p = -1.0$, and the OPD of the ray was –0.139 waves when Z = −10 mm and $y_p = 1.0$, as shown in Fig. 2(d). When Z = 10 mm and $y_p = -1.0$, the OPD of the ray was −0.110 waves, and when Z = 10 mm and $y_p = 1.0$, the OPD of the ray was 0.089 waves, which indicated the asymmetry of edge rays was optimized. Table 1 shows some of the parameter changes in Figs. 1 (without CL) and 2 (with CL) c and d. The OPD ($x_p/y_p = \pm 1.0$) of the edge rays decreased and converged to the same value in the case with CL compared to the case without CL, which indicated that the symmetry of the rays was improved in the Z = 10 mm and Z = −10 mm planes. The STRH within the 3D scanning range after correction is shown in Fig. 2(b). Due to the increase in beam symmetry and the decrease in the OPD between the edge and chief rays, the spot uniformity was significantly improved. The STRH of the scanning field corners of the system remained above 0.5 in a 3D scanning range of $40 \times 40 \times 20\ mm^3$, and the volume of the area with STRH of ≥0.8 in the scanning range was expanded by 15.2%.

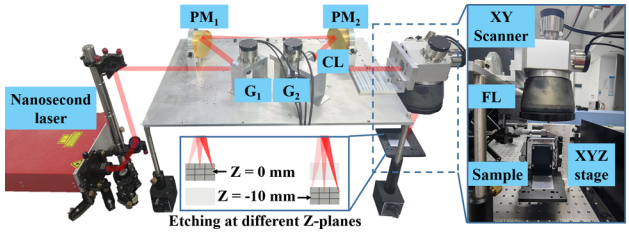


Fig. 3. Experimental setup of an RDFD-based laser scanner with ASC. Laser, nanosecond fiber laser; G_1, G_2, galvanometers; PM_1, PM_2, off-axis parabolic mirrors; FL, F-theta lens; XY scanner, two-axis galvanometers; CL, aspheric correcting lens.

An experiment was carried out to verify the effectiveness of the ASC, and the experimental setup is shown in Fig. 3. A nanosecond fiber laser (YDFLP-100-M8-M-R, JPT) with a 3.0 mm (expanded to 8 mm, for observing the experimental results more dramatically) beam diameter was adopted for the experiment. The wavelength was 1064 nm, the beam quality factor M^2 was 1.4, and the maximum average power was 100 W. The curvature R of the off-axis parabolic mirrors (35588, Edmund) was 304.8 mm, and the offset angle was 45°, corresponding to the initial state of G_1. The G_1 and G_2 can be deflected by ±12.5°. The scanning speed of the XY scanner (ExtraScan II 14, Hans) was up to 7 m/s. An F-theta lens with a focal length of 100 mm was used to focus the laser, and the working distance was 142 mm. An XYZ stage was placed to move the samples. Two typical planes of Z = 0 mm and Z = −10 mm were selected to evaluate the effectiveness of ASC. An aluminum (1060) plate with a size of $50 \times 50 \times 1\ mm^3$ was used for experiments, and the laser etching effect was used as the criterion for evaluating the effective scanning range. The laser scanning path was a $40 \times 40\ mm^2$ rectangle, with horizontal and vertical crosshatched lines spaced at 0.5 mm apart. The experimental laser energy was 100 μJ, the laser frequency was 85 kHz, and the scanning speed was 2 m/s. A laser scanning confocal microscope (VK-X1100, Keyence) was used to observe and compare the scanning effect. Table 2 shows the parameters of the CL. It was a positive lens with an even-ordered aspherical surface at the incidence and a standard surface at the exit.

Figure 4 shows the STRH simulation and the laser etching results at two dynamic focusing planes. At the Z = 0 mm plane, the STRH was close to 1.0 in the range of $40 \times 40\ mm^2$ without or with the ASC method, as shown in Figs. 4(a) and 4(e). Complete and uniform laser-etched rectangles are shown in Figs. 4(c) and 4(g). The size of laser etching points within the $40 \times 40\ mm^2$ scanning range was uniform, the etching effect was basically the same, and the depth profiles were up to 10 μm, as shown in Figs. 4(b), 4(d), 4(f), and 4(h).

Table 2. Parameters of the CL

Surface	Radius	Thickness	Glass	Conic	2nd Order
Even Asphere	57.008	3.596	F_silica	-0.191	4.936E-4
Standard	57.014	63.438	/	0.000	0.000

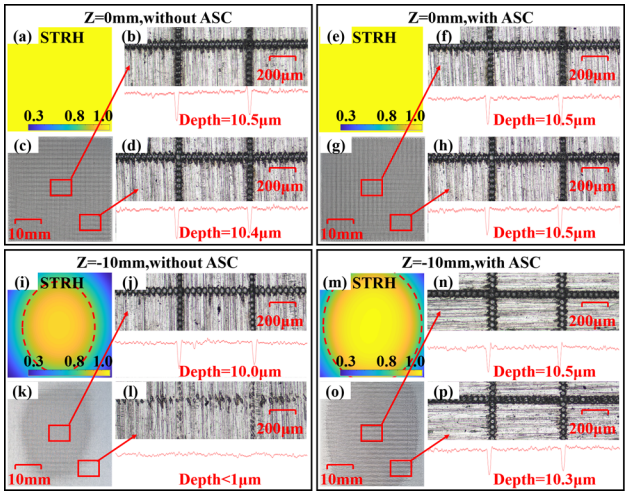


Fig. 4. STRH simulation and scanning microscopic observations of the system at two dynamic focusing planes. (a)–(d) Z = 0 mm, without ASC; (e)–(h) Z = 0 mm, with ASC; (i)–(l) Z = −10 mm, without ASC; (m)–(p) Z = −10 mm, with ASC.

At the Z = −10 mm plane, without the ASC method, the STRH decreases, and the region with STRH of ≥0.8 was only an ellipse in the 40 × 40 mm^2 area based on the STRH of the regional center of 0.93, as shown in Fig. 4(i). The laser etching result is shown in Fig. 4(k), where the laser etching lines could be only observed at the center region. However, the size of the laser-etched points was reduced, as shown in Fig. 4(j). At the edge region, the laser etching lines became less visible and did not form a complete laser-etched point, and the depth profile was less than 1 μm, as shown in Fig. 4(l). After correction, the region with STRH of >0.8 was expanded based on the STRH of the regional center increasing to 0.99, as shown in Fig. 4(m). The effective laser etching region was also expanded, as shown in Fig. 4(o). The etching quality of both the center and edges of the effective scanning region was also improved, and the depth profiles were up to 10 μm, as shown in Figs. 4(n) and 4(p). The above results show that the ASC method extended the effective scanning range of RDFD-based laser scanners.

In conclusion, the off-axis aberrations in RDFD-based laser scanners were presented and discussed. Due to the disruption of beam symmetry and the increase in the OPD between the edge and chief ray, the off-axis aberrations were introduced, and the 3D scanning range of RDFD-based laser scanners was reduced. An ASC method was proposed to optimize the aberration and extend the scanning range. The convex lens with an aspherical surface improves the symmetry of the beam and reduces the OPD between the edge rays and the chief ray. The effectiveness of the ASC method was confirmed by laser etching experiments, and the 3D scanning range of the RDFD-based laser scanner has been increased by 15.2%. The ASC method has the potential to be applied in all RDFD-based laser scanners or other off-axis systems.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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